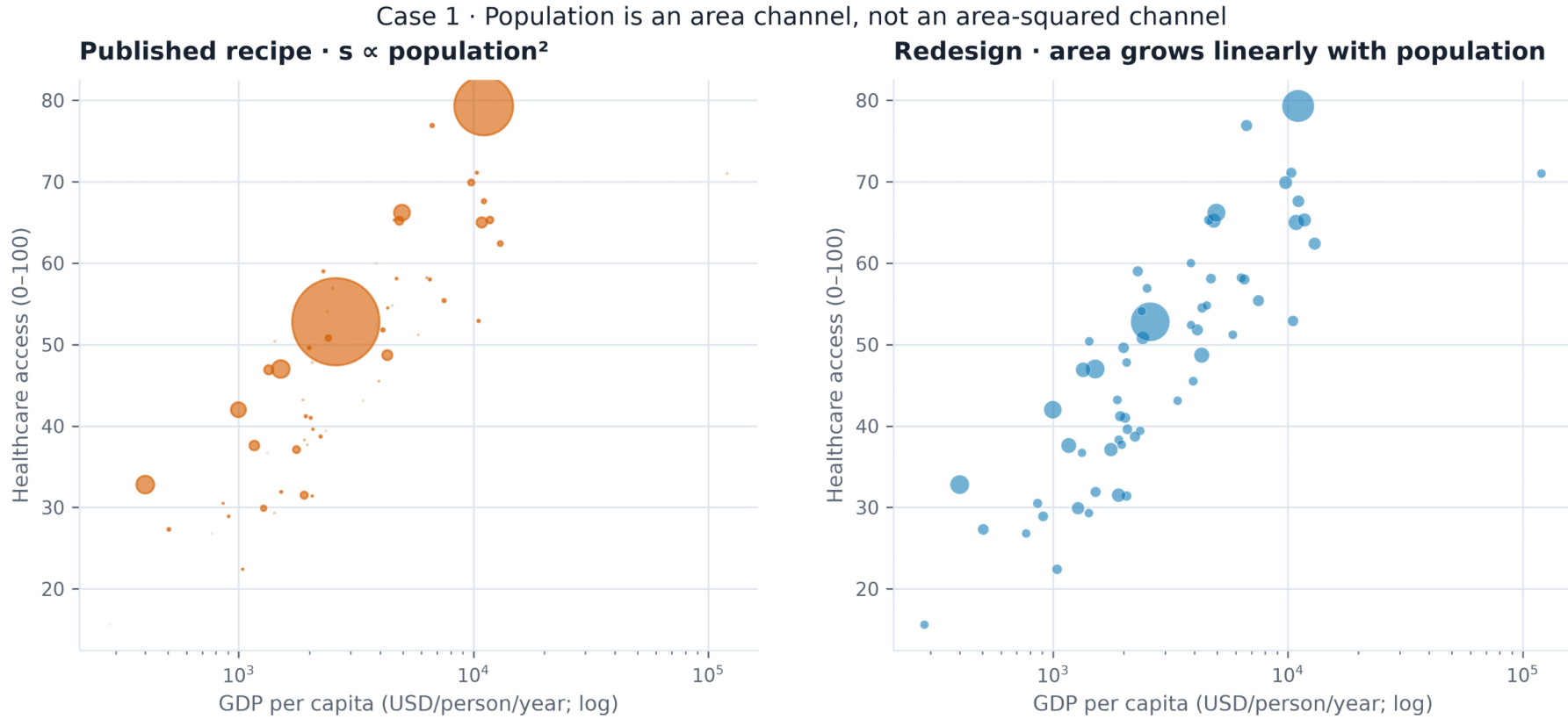


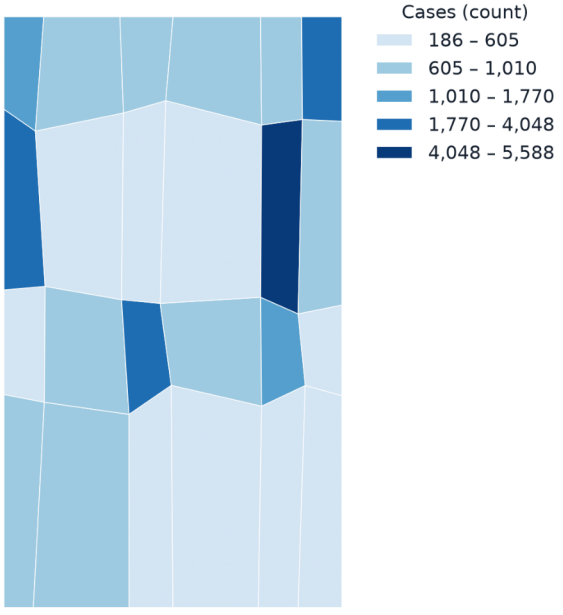

Synthetic health_indicators.csv; 60 countries; marker areas in points².

Case 1 — population squared into scatter area

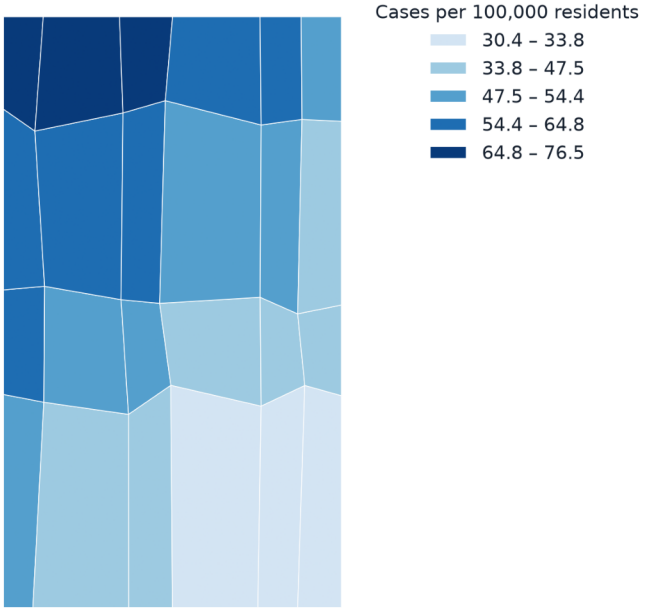
- Invited conclusion: Population and healthcare access appear unrelated across the GDP range.
- Decoy ruled out: There are 60 marks, and GDP's log scale is correct; the issue is not missing countries or the x transform.
- Mechanism: Matplotlib's s already means area. Setting s=pop_m2 makes area quadratic in population and reduces small-country marks to near invisibility.
- Evidence: Population spans 0.6–353.4 M (589×); the supplied recipe implies an area ratio of 346,921× (589²), and 9 countries are below 2.0 M. The brief's stated 589× area check conflicts with its own s=pop_m2 recipe; the code/data imply 589², so this report preserves the mathematical result.
- Redesign: Use a bounded, declared linear mapping from population to marker area; keep log GDP, retain and label the outlier, and include a size legend with real population values.

Case 2 · Counts locate population; rates locate relative incidence

**Published: raw case count**




**Redesign: population-normalized rate**

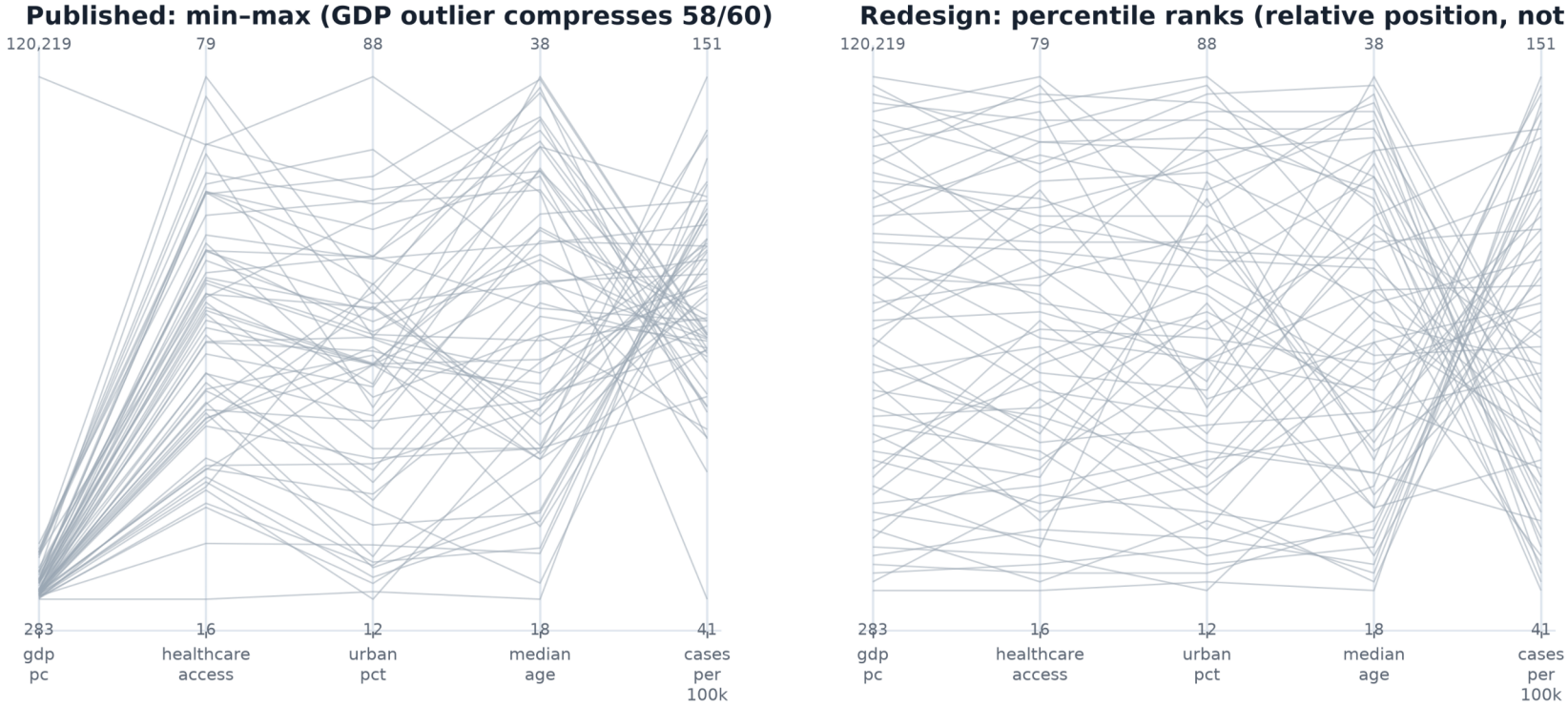



Synthetic district data and fictional district boundaries; equal-area projection; 24/24 join.

Case 2 — raw-count choropleth

- Invited conclusion: The three darkest/highest-count districts are the outbreak hotspots.
- Decoy ruled out: All 24 districts join and the count data are arithmetically correct.
- Mechanism: A choropleth's polygon area is part of the visual encoding, but the count is unnormalized; districts with more residents tend to have more cases even at equal incidence.
- Evidence: Count leaders: D17, D13, D09; rate leaders: D19, D20, D21. District counts correlate with population at r=+0.986; the count and rate top-three sets overlap 0 districts.
- Redesign: Map cases per 100,000 residents with equal-area geometry, print breaks and legend denominator, and pair the map with a count ranking.

Case 3 · The scaled GDP axis is almost flat, not the underlying data




Synthetic health indicators; min–max and percentile-rank methods named; all five non-size numeric axes.

Case 3 — min–max flattening

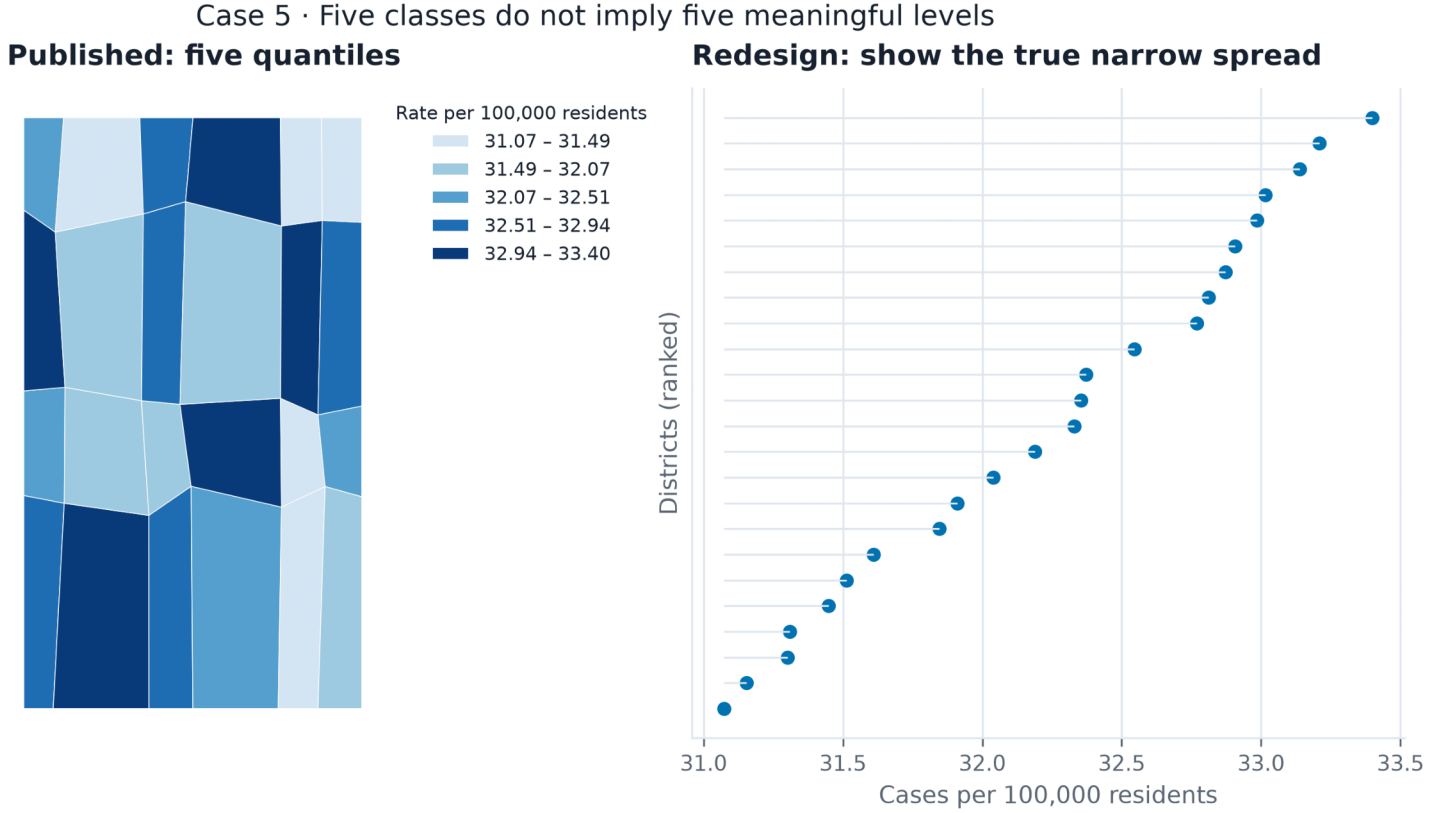
- Invited conclusion: GDP per capita barely varies across the sample.
- Decoy ruled out: There are no missing countries; the 60 GDP values genuinely span a large range.
- Mechanism: Min–max scaling divides every gap by the full range; the single top value (9.2× the runner-up) stretches the range and compresses the other observations.
- Evidence: After min–max scaling, 58/60 countries lie below 0.10 and the median is 0.017; raw GDP runs 283–120,219.
- Redesign: Use percentile ranks when the question is relative ordering (and label that magnitude is discarded), or show raw GDP on a log scale when absolute gaps matter.




Synthetic health indicators; min–max scale; file order compared with geolib.corr_chain.

Case 4 — misleading axis order

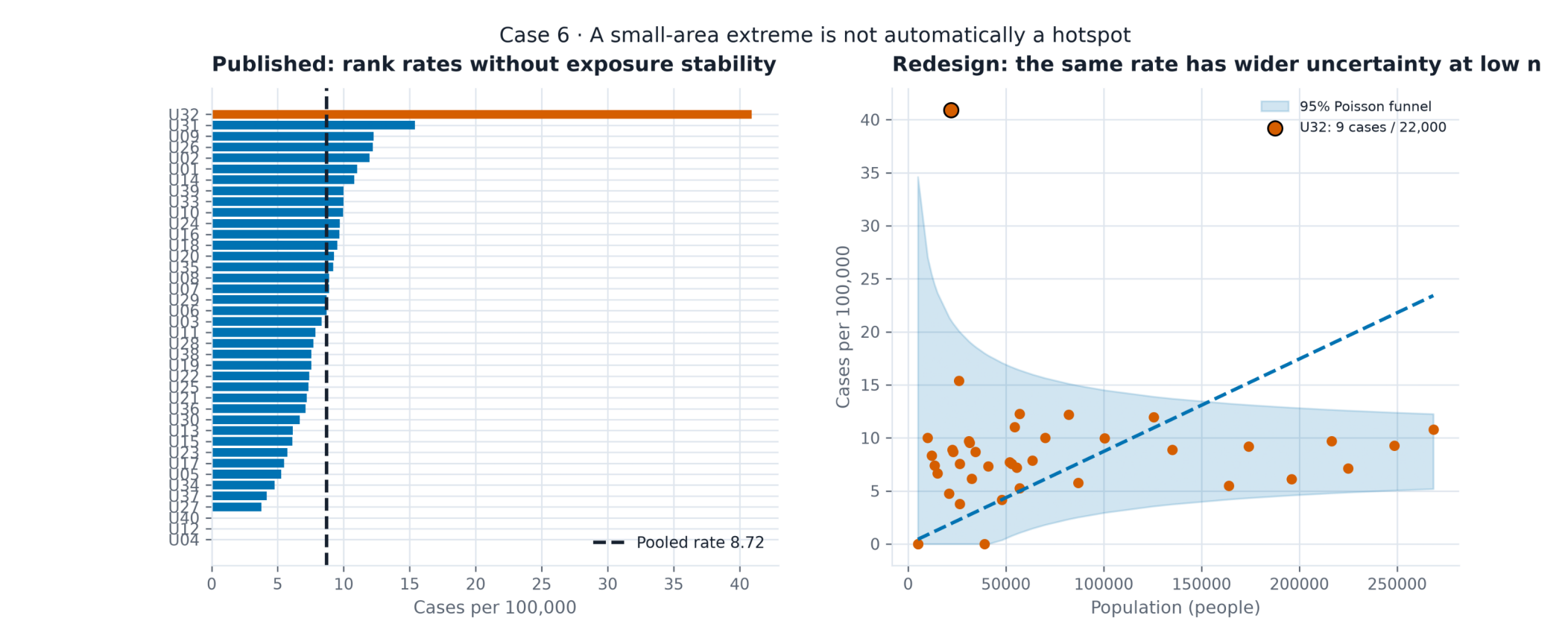
- Invited conclusion: Healthcare access and case rate appear unrelated because no crossing is obvious.
- Decoy ruled out: The five axes and the min–max scaling are valid; no scale was omitted.
- Mechanism: Parallel-coordinate crossings depend on adjacent axis pairs; the file order inserts two unrelated variables between strongly related access and incidence axes.
- Evidence: File order: gdp_pc, healthcare_access, urban_pct, median_age, cases_per_100k; greedy chain: healthcare_access → cases_per_100k → urban_pct → median_age → gdp_pc. Healthcare access and cases_per_100k have r=-0.799 but are three positions apart in the published chart.
- Redesign: Order axes using the stated greedy absolute-correlation chain; the paired variables become adjacent and the crossing pattern can be read.




Synthetic 24-district rates; quantile breaks=[31.07, 31.49, 32.07, 32.51, 32.94, 33.4].

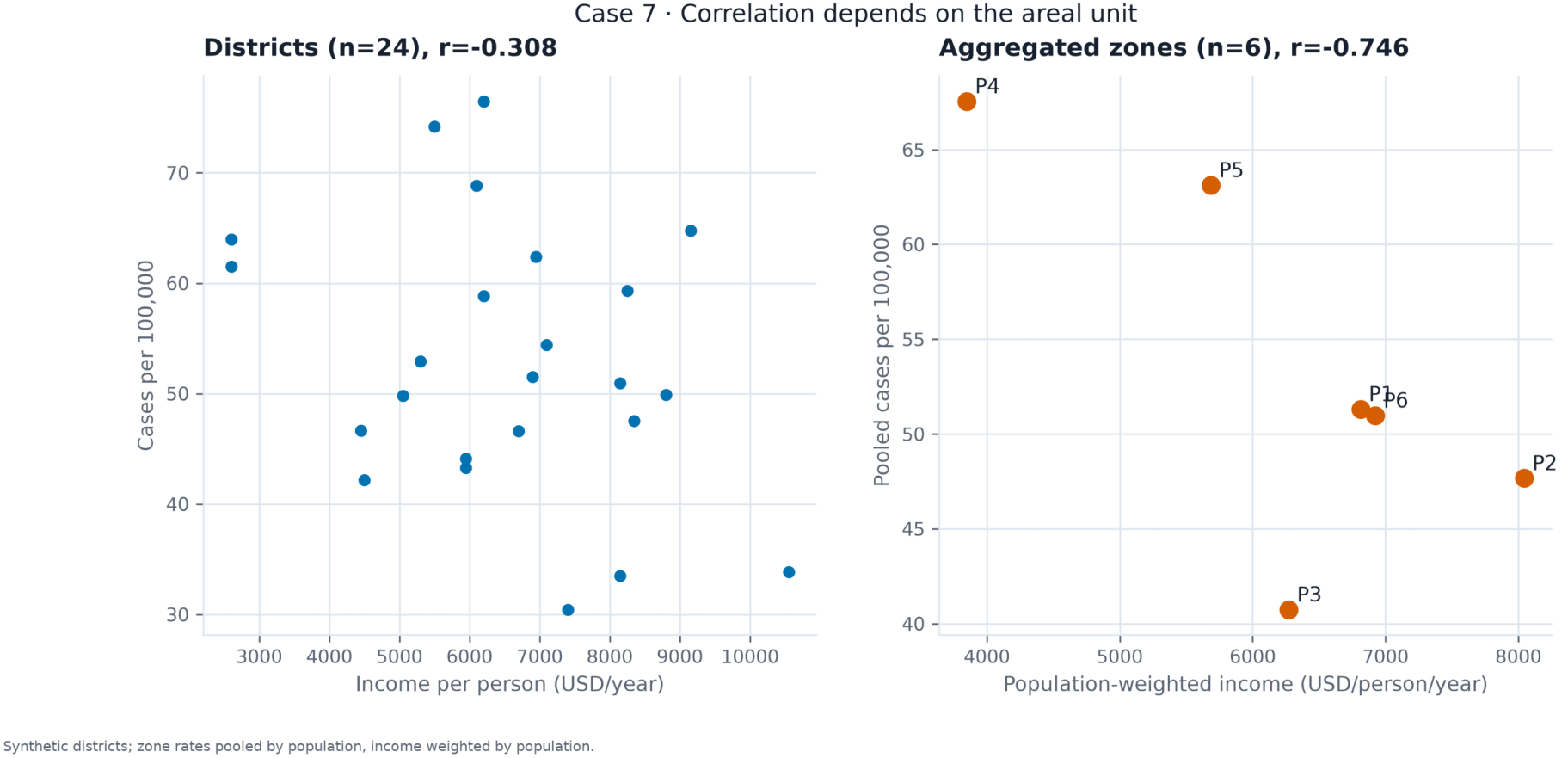
Case 5 — quantile classes over a flat distribution

- Invited conclusion: A five-level incidence gradient is visible across the federation.
- Decoy ruled out: The calculations and quantile assignments are correct, and all 24 districts are included.
- Mechanism: Quantile classification forces equal-count bins even when the underlying range is substantively tiny; class contrast exaggerates differences.
- Evidence: Rates span 31.07–33.40 per 100,000, only 7.2% of the mean. The five quantile breaks are [31.07, 31.49, 32.07, 32.51, 32.94, 33.4].
- Redesign: Show the actual values on a common linear dot scale and state that the total spread is small; use quantiles only when equal-count ranking is the explicit purpose.




Case 6 — unstable rate from a small denominator

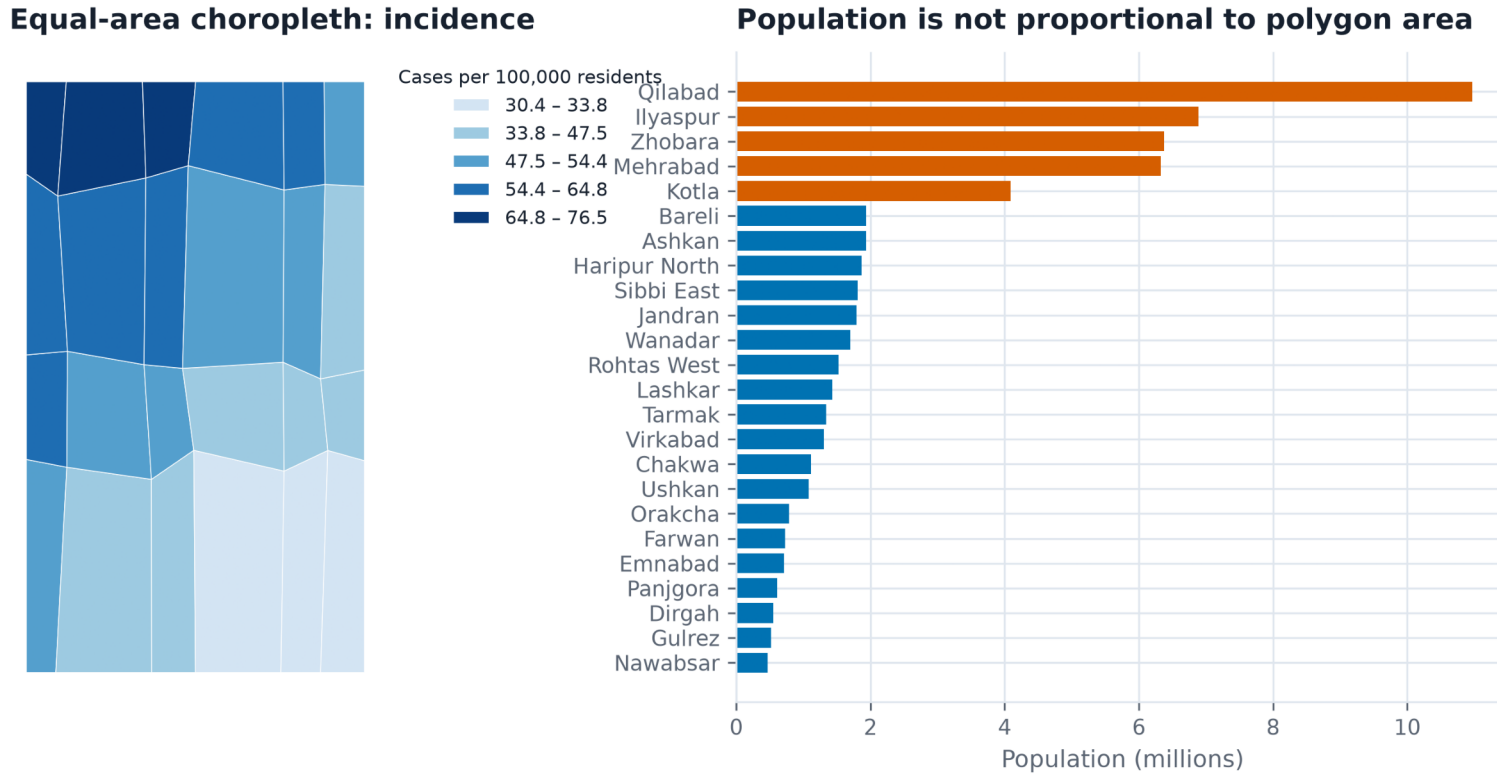
- Invited conclusion: U32 is nearly five times the national rate and therefore a hotspot.
- Decoy ruled out: The division is correct; the defect is not arithmetic or a map projection.
- Mechanism: A rate based on a small population has high Poisson variability; treating a noisy small-area estimate as a stable signal overstates certainty.
- Evidence: Pooled rate 8.72/100,000; U32 is 40.9/100,000 from 9 cases among 22,000 people (4.7×); 1 of 40 units fall outside the 95% funnel.
- Redesign: Show exposure against rate with a widening Poisson funnel and report the small-area numerator and denominator; do not rank a noisy extreme as a proven hotspot.




Case 7 — areal-scale dependence

- Invited conclusion: District income and incidence have only a weak negative relationship (r≈−0.31).
- Decoy ruled out: The district-level Pearson correlation is correctly computed across 24 complete records.
- Mechanism: A correlation is conditional on its areal units; grouping districts into zones changes within-zone variation and can produce the modifiable areal unit problem (MAUP).
- Evidence: District r=-0.308; reaggregated population-weighted zone income versus pooled zone incidence r=-0.746 across 6 zones. These are different estimands, not evidence that one correlation is universally true.
- Redesign: Show the district scatter and report zone aggregation as a sensitivity analysis, with the unit of analysis named.

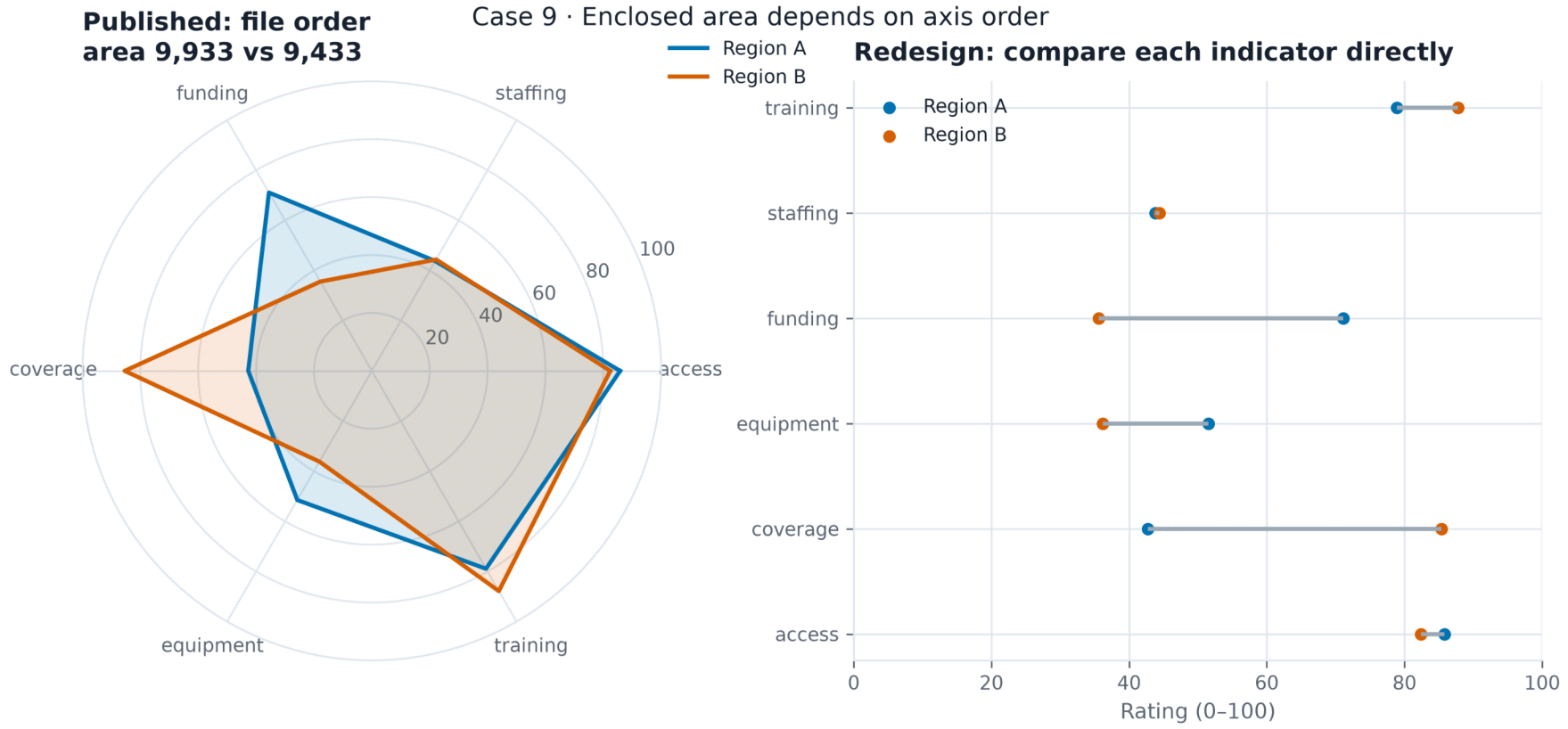
Case 8 · A dark land area does not mean most people live there




Synthetic 24-district data; equal-area projection; top-five residents=59.9%, land=12.6%.

Case 8 — area-weighted visual impression

- Invited conclusion: The outbreak is rural because much of the map is dark.
- Decoy ruled out: The rate is normalized, the sequential scale is correct, and the map uses equal-area geometry.
- Mechanism: A choropleth gives every square kilometre visible area regardless of population; viewers infer prevalence from the amount of colored land instead of the share of residents affected.
- Evidence: The five most populous districts contain 59.9% of residents but only 12.6% of land area; dark-map area therefore does not measure people or the rural share.
- Redesign: Pair the rate map with population-weighted counts/rankings or proportional symbols; explicitly describe population rather than colored land area.




Synthetic six-indicator profiles; means differ by 0.33 points.

Case 9 — radar area depends on order

- Invited conclusion: Region A has a visibly stronger profile because it encloses more area.
- Decoy ruled out: Both profiles use the same 0–100 scale and the same six indicators.
- Mechanism: Radar polygon area depends on the adjacency/order of axes; a rearrangement changes cross-products and enclosed area without changing any score.
- Evidence: Mean scores are 62.3 and 62.0 (difference 0.33); in supplied order their polygon areas are 9,933 and 9,433, a layout-dependent difference.
- Redesign: Use paired points or sorted dumbbell bars on a shared linear scale to compare each indicator; do not summarize a radar polygon's area as overall strength.